

Sonika Singh Tomar

B.Tech CSE 📞 +91-7000565895

✉ tomarsonika10@gmail.com

🌐 linkedin.com/in/sonika-tomar-760048199

🐙 github.com/testgithubsonika

Education

VIT Bhopal University

2023 – 2027

Bachelor of Technology in Computer Science Engineering (CGPA: 8.56/10.0)

- **Academic Performance:** Class XII – 90.6%, Class X – 92.2%
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

Technical Skills

Languages: Python, JavaScript, C++, SQL, HTML/CSS

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, Tailwind CSS

Machine Learning: PyTorch, XGBoost, scikit-learn, SHAP (Explainable AI)

Databases & Caching: PostgreSQL, Redis, Sequelize ORM

Cloud & DevOps: AWS, Docker, Git, GitHub

Concepts & Tools: REST APIs, WebSockets, JWT Authentication, Google OAuth, Postman

Projects

AI-Powered Train Ticket Booking System

Mar 2025 – Aug 2025

React.js, Node.js, Express.js, PostgreSQL, FastAPI, Redis, Tailwind CSS

Live – GitHub

- Built a full website where users can search for trains, book tickets, cancel bookings, and view their past bookings.
- Added an AI feature that predicts how likely a ticket is to get confirmed, using FastAPI and XGBoost.
- Built the backend APIs and database using PostgreSQL, so the app can handle many users and bookings smoothly.
- Used Redis for faster performance, background job processing for tasks like sending confirmations, and secure login with JWT and Google sign-in.

Sentinel – AI-Powered Behavioral Anomaly Detection System

Feb 2026 – April 2026

React.js, FastAPI, PostgreSQL, PyTorch, XGBoost, WebSockets, Tailwind CSS

Live – GitHub

- Built a security monitoring system that detects suspicious user and device behavior, like brute-force logins, impossible travel, and unusual access patterns, using machine learning.
- Combined three ML models (Isolation Forest, a Bi-LSTM neural network, and XGBoost) to catch unusual behavior and correctly identify the type of attack in real time.
- Added an explainability layer using SHAP so every alert shows a clear reason (e.g. "unusual login location + failed logins"), instead of just a raw score.
- Built a FastAPI backend with a live WebSocket connection and a React dashboard that updates instantly as new threats are detected, with no page refresh needed.

Achievements

- Codeforces Specialist (Max Rating: 1420)
- AWS Certified Solutions Architect – Associate (2026)
- AWS Certified Developer – Associate (2026)
- Google IT Support Professional Certificate (Coursera)
- Solved 500+ Data Structures & Algorithms problems on LeetCode